

Christopher Kitras

chkitras@gmail.com

kitras.dev

linkedin.com/in/christopher-kitras

EDUCATION

Brigham Young University – Fulton College of Engineering

- PhD Electrical and Computer Engineering (3.97/4.00)
- BS Computer Engineering (3.6/4.0)

Provo, UT
April 2027
December 2021

COMPUTER/TECHNICAL SKILLS

- Python
- C/C++/CUDA
- Linux/Bash
- Systems Engineering
- Computer Networks
- AVR/ESP
- InfluxDB/Grafana
- SystemVerilog/Verilog

ENGINEERING EXPERIENCE

Project: Latency-based WiFi Subprotocol Development April 2024 – July 2026

Objective: Created a side channel that can infiltrate/exfiltrate information in/out of a network protected by classic security and segmentation techniques

- Engineered a covert-channel transmitter by repurposing a Linux kernel packet-generation module to sustain traffic bursts exceeding 1,000,000 packets/sec
- Created a robust receiver that infers router disruption from 3 interchangeable proxy signals
- Validated feasibility across 6 commercial router models via controlled test campaigns
- Achieved a 95% packet reception ratio (PRR) over a real-world WAN path spanning 14,000 km, 300+ ms latency, and high jitter
- Authored a paper (under review) presenting the attack and discussing potential mitigations

Project: Jitter-based WiFi Subprotocol Security Audit July 2024 – April 2025

Objective: Audited unconventional protocol for vulnerabilities and implemented security measures

- Identified attack vectors where an eavesdropper could sniff and replay data
- Implemented a lean message authentication code (MAC) that ensures data integrity by hashing the message, a shared secret, and a timestamp
- Collaborated with a joint team to devise a strategic implementation for an optimized solution
- Authored a manuscript section for Jitter-based Subprotocol explaining and justifying the proposed security measures

Project: Traffic Camera Air Quality Monitoring Platform April 2023 – April 2025

Objective: Developed a platform that enables ML researchers to access to large datasets to enable real-time model training

- Collected a dataset over 3TB full of traffic camera images around Utah
- Created new and utilized pre-existing APIs to gather available environmental data
- Designed and maintained web app using MongoDB and Flask instance to store and deliver data
- Integrating a pre-existing ResNet50 model that will create a regression between available PM2.5 values and the traffic camera images

LEADERSHIP

- Lead Research Student, IoT Resilience, Network Enhanced Technologies Lab
- President Emeritus and Officer, Brigham Young University Linux Club
- Coordinate welfare and administrative operations in a local immigrant congregation

WORK HISTORY

- *PhD Candidate, IoT Systems Resilience Research, Brigham Young University* 2022 – Present
- *Cybersecurity Research Intern, Subprotocol Discoverey, Riverside Resarch* 2025
- *Instructor, ECEN 224 Lab and Recitation, Brigham Young University* 2023
- *Research Assistant, Firmware and GUI development, Brigham Young University* 2019 - 2022
- *Instructional Aide, Fundamentals of Digital Systems, Brigham Young University* 2019 - 2020